

FairRSFM: A Biome-Aware Benchmark and Debiasing Framework for Remote Sensing Foundation Models



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Subject Area: Remote Sensing, Earth

Observation, and Sensors

Student Paper: Yes

Track: Evaluations and Datasets Track

Abstract:

Remote sensing foundation models (RSFMs) are commonly evaluated using aggregate metrics, which can hide systematic performance disparities across ecological regions. We introduce FairRSFM, a biome-aware benchmark for evaluating ecological group robustness in RSFMs. FairRSFM augments four downstream datasets, m-EuroSAT, m-BigEarthNet, m-SA-Crop-Type, and MMEarth20K with Dynamic World label maps, with biome metadata derived from georeferenced samples. Each sample is mapped from 14 terrestrial biome classes into six ecologically meaningful macro-groups, enabling standardized worst-group and cross-biome disparity evaluation across classification and segmentation tasks. Using Prithvi-EO-2.0 under a unified frozen-backbone linear-probing protocol, we show that aggregate performance masks substantial biome-dependent disparities. For example, m-EuroSAT reaches only 0.836 macro-F1 in the Aseasonal High-Biomass group, while m-SA-Crop-Type drops from 0.289 to 0.176 mIoU between Mediterranean regions and Deserts/Xeric Shrublands. As an illustrative mitigation baseline, Biome-Orthogonal Linear Probing improves worst-group performance from 0.836 to 0.843 on m-EuroSAT and from 0.495 to 0.520 on m-BigEarthNet without updating the RSFM backbone. FairRSFM provides a reusable benchmark and protocol for measuring ecological robustness gaps in remote sensing foundation models.

Submission Number: 2158

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Meta Review of Submission2158 by Area Chair wFd7

Meta Review by Area Chair wFd7 31 Jul 2026, 07:15 (modified: 10 Aug 2026, 10:35)
Senior Area Chairs, Area Chairs, Program Chairs, Authors, Reviewers Submitted Revisions

Metareview:

This paper introduces a biome-aware evaluation framework for remote-sensing foundation models by attaching ecological-region metadata to four existing datasets and reporting subgroup robustness metrics. It also evaluates Dynamic Biome Reweighting and Biome-Orthogonal Linear Probing. Reviewers agree that ecological subgroup robustness is important, underexplored, and potentially useful to the geospatial community.

However, the empirical evidence is not mature enough for a benchmark or datasets-track contribution. Necessary changes in the next cycle include broader model coverage, repeated experiments, stronger baselines, and substantial benchmark-design validation.

Final Recommendation: Recommend resubmission in round 2 with updated paper and rebuttal

reviewed

Official Review by Reviewer 6Aow 25 Jul 2026, 02:35 (modified: 10 Aug 2026, 10:19)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 6Aow, Authors Revisions

Paper Summary:

This paper proposes FairRSFM, a biome-aware benchmark and debiasing framework for auditing whether remote sensing foundation models perform consistently across ecological regions rather than only well on average, as well as DBR and BOLP, two methods for debiasing a model. The motivation for this work is that aggregate performance metrics (accuracy, macro-F1, mIoU) can hide large performance gaps between different biomes. To evaluate this, the authors assign a biome to each sample in existing datasets (m-EuroSAT, m-BigEarthNet, m-SA-Crop-Type from GEO-Bench, plus a 20K-sample MMEarth subset with Dynamic World labels) via the centroid coordinates and an external terrestrial-ecoregion map. They then use this to compute standard fairness metrics (worst-group score, a normalized range "NFR," group std-dev, and equalized-odds/demographic-parity gaps) over the biome sub-groups.

The experiments consist of linear probing Privthvi-EO-2 using "ERM", as well as their proposed BOLP and DBR methods. The results show that strong aggregate scores do mask real biome-dependent gaps (e.g., m-EuroSAT worst group 0.836 macro-F1 vs. 0.940 overall; m-SA-Crop-Type 0.176 mIoU on Deserts/Xeric vs. 0.290 on Mediterranean), that BOLP yields small worst-group improvements on the two classification tasks (m-EuroSAT 0.836 to 0.843; m-BigEarthNet 0.497 to 0.520) while sometimes hurting segmentation (it degrades MMEarth20K), and that naive biome-balanced sampling actually lowers performance without helping the worst group.

Paper Strengths:

- Well-motivated, under-explored problem. Ecological/biome conditioned robustness is a largely unaddressed component of remote-sensing model fairness. Spatial generalization is often discussed, however distant locations may have similar biomes and therefore look more similar than nearby locations in different biomes.
- BOLP is really interesting and I'd love to see more results using it.

Major Weaknesses:

- There are no confidence intervals or repeated runs, which makes it impossible to draw conclusions from the results. All numbers in Tables 4-6 come from single runs with no error bars or significance tests, yet the claimed effects are tiny relative to the sampling noise. For example, the m-EuroSAT worst-group gain of +0.007 that is pointed out in the abstract is computed on **just** 25 test samples, where the binomial standard error alone for this would be $\sim \pm 0.14$ (i.e. 20x larger).
- Detail oriented pass needed. E.g. "On m-EuroSAT, BOLP improves the worst group score from 0.800 to 0.843", but from Table 6 it looks like it improves from 0.836 to 0.843. E.g. abstract cites m-BigEarthNet worst-group 0.495 to 0.520, but Tables 5 and 6 report 0.497 to 0.520.
- NFR and std don't make sense for m-SA-Crop-Type as there is only 2 groups here. E.g. standard deviation of 2 numbers (a and b) is just $|a-b|/2$. It is a stretch for the m-EuroSAT, and m-BigEarthNet ones as well, as they only have 3 groups.

Minor Weaknesses:

- L16, "m-EuroSAT reaches only 0.836 macro-F1 in the Aseasonal High-Biomass group" -- 0.83 macro-F1 seems high?
- L104 the "ERM" acronym is never defined. I'm assuming empirical risk minimization.

Preliminary Recommendation: 2: Weak Reject

Preliminary Justification:

I think the overall idea is interesting; however, the results need to have confidence intervals and/or repeated runs in order to draw conclusions from them. Given the main experiments here are linear probes over fixed embeddings, this isn't a huge computational demand.

Resubmission: No, the required revisions are too extensive for a resubmission in the time that is available.

Revisions For Resubmission:

n/a

Confidence Level: 4 - High Confidence: The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

Addresses an important challenge, but dataset coverage and experiments are limited

Official Review by Reviewer F5hw 13 Jul 2026, 23:35 (modified: 10 Aug 2026, 10:19)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer F5hw, Authors [Revisions](#)

Paper Summary:

This paper presents FairRSFM, a benchmark dataset for evaluating performance disparities across biome subgroups for remote sensing foundation models. The benchmark adds biome metadata (14-class biome groups + a macro-grouping) to four datasets (m-EuroSAT, m-BigEarthNet, m-SA-Crop-Type, and MMEarth20K) and evaluates Prithvi-2.0 in a linear probe setting. The paper also evaluates several debiasing approaches including standard ERM, Dynamic Biome Reweighting (DBR), and Biome-Orthogonal Linear Probing (BOLP). Results show that worst-group performance is substantially lower than average performance for Prithvi on all datasets.

Paper Strengths:

- The paper addresses the problem that aggregate metrics mask biases or performance disparities across geographic subgroups. This has far-reaching consequences in geospatial AI but there are no strong benchmarks that tackle this.
- The paper is easy to read and follow technical details. Overall the figures and tables are neat and well-formatted.
- The experiments evaluate multiple debiasing techniques, including naive sample-balancing, to evaluate how well existing approaches can correct for bias resulting from the dataset distributions.

Major Weaknesses:

- Is it a case study, a benchmark, or an evaluation/insights paper? The paper seems to try to toe the line between claiming impactful contributions but not overstating the generality of the findings. I feel this is because the experiments shown in the paper are not complete enough for this paper to be accepted as is.
 - The choice of four datasets is not explained and feels unnecessarily limited. Why were these four chosen? Are there known biases in them that can provide good test cases? Adding biome metadata to georeferenced datasets is very simple, so it seems unnecessarily limiting to choose four (unless those four are particularly informative where others are not).
 - The paper only evaluated with Prithvi-2.0 and no hyperparameter tuning. The paper writes that Prithvi is a case study and other models can be run with it - then I would expect this paper to do that. Given the linear probe only setting and no HP tuning, it this should not be very costly.
 - The paper does not report any results with confidence intervals/standard errors. This seems especially important given the small size of some of the test sets (as small as 14-25 points as shown in table 5). It would be helpful to see the group-wise counts in the train/val/test sets for each dataset not just MMEarth20K (appendix would be fine, but this is more important information than table 2 in my opinion)

Minor Weaknesses:

- There is a lot of paper-space spent on explaining the macro-groups which feels like an unnecessary emphasis compared to other information (like sample counts or a map of biomes/macrogroups, which would help non-RS readers).
- L86: "To address this gap..." The paragraph before does not talk about a gap, so this seems like a revision artifact.

Preliminary Recommendation: 3: Borderline Reject

Preliminary Justification:

I think the paper addresses an important problem and provides well-organized experiments and results, but the experiments are not complete enough in their current form. To move towards acceptance, I would need the authors to address the limited scope/lack of motivation of the datasets, add more models, and add standard errors to address the low-sample sizes in the test sets.

Resubmission: Yes, the paper is of reasonable quality but requires some revisions before it can be accepted.

Revisions For Resubmission:

To move towards acceptance, I would need the authors to address the limited scope/lack of motivation of the datasets, add more models, and add standard errors to address the low-sample sizes in the test sets.

Confidence Level: 5 - Expert: The reviewer is an authority in the specific subfield addressed by the paper. They are extremely confident in their evaluation and understanding of the work.

review

Official Review by Reviewer sdEa 13 Jul 2026, 18:42 (modified: 10 Aug 2026, 10:19)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer sdEa, Authors [Revisions](#)

Paper Summary:

This paper introduces FairRSFM, a benchmark for evaluating ecological robustness of RSFM. Instead of reporting only aggregate performance metrics, the benchmark augments four existing remote sensing datasets with biome metadata and proposes evaluating performance across biome derived groups. The paper also introduces two simple mitigation strategies under a frozen-backbone linear probing protocol. The main message is that aggregate metrics mask substantial performance disparities across ecological regions.

I find the motivation timely and relevant. Robustness and fairness of foundation models are increasingly important topics, and ecological subgroup evaluation is an interesting perspective. However, I am less convinced that the work currently reaches the level expected of a top-tier dataset track paper. The benchmark primarily consists of adding metadata to existing datasets and evaluating a single RSFM, while the proposed mitigation methods are relatively lightweight adaptations of existing group robustness and representation debiasing techniques. Consequently, it is unclear whether the paper provides sufficient novelty or sufficient empirical depth to justify publication in its current form.

Paper Strengths:

The paper addresses an important and underexplored problem: ecological subgroup robustness in remote sensing foundation models.

The benchmark is practically useful, as biome metadata can be reused for future evaluation of RSFMs.

The paper demonstrates that aggregate metrics can hide large subgroup performance disparities, which is an important observation.

The evaluation protocol is consistent through frozen-backbone linear probing, making comparisons straightforward.

The proposed benchmark could become useful for future work if broadly adopted.

Major Weaknesses:

For a dataset-track submission, the contribution largely consists of attaching biome labels to existing benchmarks, defining evaluation metrics, and proposing two mitigation baselines. The datasets themselves are not new. This raises the question of whether the work represents a sufficiently substantial benchmark contribution compared with existing benchmark papers such as GEO-Bench.

Although the paper motivates benchmarking remote sensing foundation models, essentially every experiment is performed using only Prithvi-EO-2.0. A benchmark should ideally characterize an ecosystem of models rather than one case study. This is needed for generalisation.

The paper merges 14 biome classes into 6 macro-groups. However there is little justification that these six groups are optimal. A few questions remain like why six groups? why these merges? would eight groups produce different conclusions? would the original fourteen groups be preferable?

Representation debiasing is relatively incremental.

The paper adopts Equalized Odds and Demographic Parity. These metrics originate from human demographic fairness. It is not fully convincing that they directly transfer to ecological regions. The paper could better justify why these fairness notions are meaningful in ecological settings.

For a convincing benchmark paper, stronger baselines from group robustness literature should be included. Examples include GroupDRO.

For the semantic segmentation tasks (m-SA-Crop-Type and MMEarth20K), the authors assign the entire patch to a single biome macro-group based solely on its spatial centroid (lines 213–216). While this holds for patch-level classification, it introduces severe label noise for dense pixel-level prediction. Remote sensing tiles often capture ecotones (transitional boundaries between different biomes) or complex multi-biome scenes. Assigning a whole patch's segmentation mask to one biome completely overlooks local ecological variations.

Minor Weaknesses:

BOLP assumes biome labels at inference. How realistic is this assumption?

Preliminary Recommendation: 3: Borderline Reject

Preliminary Justification:

The problem addressed by the paper is important, and the benchmark has the potential to be useful for the remote sensing community. However, in its current form, I am not fully convinced that the contribution is sufficiently comprehensive for a dataset-track publication. The benchmark is evaluated on only a single foundation model, several design choices (particularly the biome grouping) are insufficiently justified, and the proposed mitigation methods are relatively incremental adaptations of existing techniques. Strengthening the empirical evaluation with multiple RSFMs, stronger robustness baselines, and more rigorous statistical analysis would substantially improve the work.

Resubmission: No, the required revisions are too extensive for a resubmission in the time that is available.

Revisions For Resubmission:

na

Confidence Level: 3 - Moderate Confidence: The reviewer is reasonably knowledgeable about the topic. They understand the paper's methodology and results.